Bharat Singh Rawat, Ph.D.

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LinkedIn
Online Dossier
PY-BEMCS

Expert in charged particle beams with over 7 years of experience in the design of ion sources, beam diagnostics, and transport systems.

Employment

03/2026 – Present

Lead Developer / Independent Researcher

(a) Developing **PY-BEMCS**, a Python-based multiphysics suite for simulating ion beam extraction and Monte Carlo Collision (MCC) physics in Ion Sources/Thrusters and Fusion Neutral Beam Injectors (NBI).
(b) Developed a self consistent **Poisson solver** integrating Bohm criterion based extraction.
(c) Implemented dynamic **sputter erosion and structural failure logic** using probabilistic Charge-Exchange (CEX) scattering models and in-situ mesh erosion to predict grid End-of-Life (EOL).
(d) Integrated a modular multiphysics backend coupling **thermal heat maps** with secondary electron emission (SEE) and RF-based co-extraction models.
(e) Developed a **GUI and DAQ system** providing real-time visualization of plasma meniscus, damage maps, electron backstreaming potential, and beam divergence.

2023 - 2026

Research Associate: University of Liverpool / Cockcroft Institute / CERN.

Responsibilities:

(a) Led **ballistic antiproton transport optimization studies** during the annual shutdown, utilizing a **3D CST Penning Trap model** to validate the new Positronium target design and determine critical operational settings **to achieve antiproton velocities below 2000 m/s**.

(b) Carried out simulation studies on the **Rotating Wall trap** of the AEgIS apparatus for **identifying the optimal frequency** for trapping antiprotons.

(c) Developed **Reduced Order Models (ROMs)** for a **Digital Twin** of the Penning Malmberg trap apparatus, integrating first principles physics with Machine Learning (ML) models to **predict trap behavior and optimize particle confinement.**

(d) Executing **critical on shift responsibilities** during beam time, including **real-time data acquisition and the management of cryogenic systems for superconducting magnets**.

2018-2018

Radiation Protection Engineer: Nuvia India Pvt Limited.

Responsibilities:

(a) Managed occupational radiation exposure for plant personnel at Kakrapar Atomic Power Station, ensuring **strict adherence to weekly, monthly, and annual regulatory dose limits**.

(b) Developed an **automated dose tracking spreadsheet to monitor exposure trends and identify personnel approaching threshold limits**, enabling proactive safety interventions.

Education

2018 - 2024

Ph.D., Homi Bhabha National Institute / Institute for Plasma Research in Engineering Sciences.

Thesis title: *Studies on extraction of an ion beam and its transport from a multicusp gridded ion source.*

Contribution:

(a) Designed, developed and characterized a **ring cusp ion source** for space propulsion and fusion research, capable of producing Ar^+ ion beams with energies of **1–2 keV** and currents up to 100 mA. Achieved an average beam current of **0.23 mA per aperture**.

(b) Engineered a nine-channel **Parallel Wire Array (PWA)** and a **Faraday Cup array (FCA)** for real-time, in-situ monitoring of ion beam profiles.

(c) Investigated space charge neutralization using a hot filament cathode, achieving a degree of neutralization up to 100%.

(d) Utilized **CST Studio for magnetic cusp optimization** and **OPERA-3D for ion optics design**, achieving a low beam divergence of 6.5°. Conducted transient thermo-structural analysis in **ANSYS** to design a grid mounting system that **limited thermal deformation to <0.1 mm.**

2015 - 2017

M.Tech., Pandit Deendayal Petroleum University in Nuclear Engineering.

Thesis title: *Design and Development of 10 kW Faraday cup for Ion beam intensity measurement .*

Contribution:

(a) Designed, developed, and characterized a Faraday Cup to handle a **thermal load of 10 kW** and a beam current of **20–30 mA**, a significant upgrade from the existing 1000 W and 500 µA system.

(b) Engineered a **water-cooling system** capable of maintaining a maximum temperature of **350.6°C**, well below the 1085°C melting point of copper.

(c) Determined an **optimal suppression voltage** of -450 V to -500 V through **SIMION 8.0** and **LORENTZ** particle tracking simulations to prevent measurement errors caused by secondary electrons.

(d) Optimized the device geometry with a 30° cone angle, which was found to **reduce the total secondary electron yield by a factor of 1.15** compared to larger angles.

2009 - 2013

B.E., Gujarat Technological University in Mechanical Engineering.

Thesis title: *Long Chain Beamer-Braking system and troubleshooting.*

Contribution:

(a) Calculated that replacing Hardened Mild Steel with **Cast Iron reduced the temperature rise** per stop by $1.39C$ (from $18.39C$ to $17C$), significantly improving safety for heat-sensitive materials like cotton.

(b) Identified and **mitigated critical operational difficulties**, including **overheating of adjacent assemblies, brake fluid vaporization, and thermal judder**, ensuring the integrity of textile materials during the beaming process.

Project title: *Synchronizing Ice Cream cup Packaging machine.*

Contribution:

(a) Analyzed and **resolved critical underperformance issues** in a four-track pneumatic packaging machine, restoring lost production capacity caused by internal mechanical misalignments.

(b) Successfully increased production output **from 9,600 cups per hour to 11,200 cups per hour, a recovery of approximately 1,600 units/hr** that brought the machine to near-maximum design capacity.

Research Publications

1. B. S. Rawat, N. Kumar, B. Rienäcker, S. Chandran, and C. P. Welsch, "**Design studies of a Retarding Potential Energy Analyser (RPEA) for low energy antimatter experiments**," in *Proc. IBIC 25*, JACoW Publishing, Geneva, Switzerland, 2025. DOI: 10.18429/JACoW-IBIC2025-MOPMO02.

2. Rawat, B. S., Kumar, N., Ghosal, D., Aflyatunova, D., Rienäcker, B., Welsch, C. P., "**Faraday cups: principles, designs, and applications across scientific disciplines—a review**," *Instruments*, vol. 9, no. 3, 2025, ISSN: 2410-390X. DOI: 10.3390/instruments9030020.

3. Rawat, B. S.,Kumar, N., Rienäcker, B., Welsch, C.P. and others, " **3D simulation studies of mixed plasma confinement at AEgIS**," in *International Conference on Exotic Atoms and Related Topics and Conference on Low Energy Antiprotons (EXA-LEAP2024)*, 2025. DOI: 10.22323/1.480.0076.

4. Rawat B.S., "**Studies on Extraction of an Ion beam and its Transport from a Multi-Cusp Gridded Ion Source**," Ph.D. dissertation, Homi Bhabha National Institute, 2024. DOI: 10.13140/RG.2.2.14806.25929.

5. Rawat, B.S and Kumar, N and Rienacker, B and Welsch, C, "**Simulation studies for the confinement of antiprotons for the AEgIS experiment**," in *Proc. IPAC'24*, JACoW Publishing, Geneva, Switzerland, 2024, pp. 2681–2684. DOI: 10.18429/JACoW-IPAC2024-WEPR75.

6. Rawat B.S., Sharma, S. K., Prahlad, V., Choksi, B., and Baruah, U. K., "**Experimental Studies on the Roles of Space Charge Neutralization in a Ring Cusp Ion Source**," *IEEE Transactions on Plasma Science*, vol. 51, no. 8, pp. 2218–2224, 2023. DOI: 10.1109/TPS.2023.3302419.

7. Rawat, B.S., Sharma, S. K., Choksi, B., Prahlad, V., and Baruah, U. K., "**Measurement system for ion beam profiles using fixed parallel wires and Faraday cup array**," *AIP Advances*, vol. 12, no. 12, p. 125 103, Dec. 2022, ISSN: 2158-3226. DOI: 10.1063/5.0117836.

8. Rawat, B.S., Sharma, S. K., Choksi, B., Bharathi, P., Sridhar, B., Gupta, L. N., Thakkar, D., Parmar, S. L., Prahlad, V., and Baruah, U. K., "**Characterization of a Multicusp Ion Source With Two-Grid Extraction System for Studying Extraction and Transport of Ion Beam**," *IEEE Transactions on Plasma Science*, vol. 49, no. 11, pp. 3537–3547, 2021. DOI: 10.1109/TPS.2021.3121519.

9. Rawat, B.S., Vala, S., Abhangi, M., Kumar, R and Chauhan, S, "**Design and simulation of 10kW Faraday cup for ion beam current**," in *Proceedings of the 25th International Conference on Nuclear Engineering ICONE25, Shanghai, China*, 2017, pp. 2–6. DOI: 10.1115/ICONE25-67589.

Skills

Languages	Strong reading, writing, and speaking competencies for English, Native in Hindi and Gujarati. Learning Spanish (B2) and Russian (A1).
Coding	Python, Matlab, C++
Software Expertiese	CST Studio (Particle tracking, ES/EM-PIC), Solid Works, OPERA3D, SIMION, COMSOL, RF Track, ANSYS (Thermal/Structural/CFD), ASTRA, MAD-X, Geant4, G4Beamline, BDSIM, IBISIMU, TAICHI, PY-BEMCS (Developer).
Outreach	Designed interactive web games related to antimatter physics for outreach demonstrations.
	Conducted interactive demonstrations showcasing superconductivity and magnetic levitation during outreach events.
Misc.	Teaching/Tutorials (Introduction to Beam diagnostics (Co-Instructor), HPC Computing, Doctoral student supervision.

Presentations and talks

1) Invited talk titled "**Design tutorial on FC design for low energy antimatter research using CST studio.**" at CST User day, October 2025, Nottingham, United Kingdom.

2) Contributed talk titled "**Digital Twin-Enabled Modeling of Antimatter Beam Control: Insights from AEgIS to EuPRAXIA**" at EuPRAXIA-DN Camp II: Science, July 2025, Lisbon, Portugal.

3) Contributed talk titled "**Trapping Antiprotons: Modeling Penning-Malmberg Traps for AEgIS at CERN**" at SIMULIA Euronorth Regional User Meeting, June, 2025, Manchester, United Kingdom.

4) Contributed talk titled "**Design and Simulation of 10 kW Faraday Cup for Ion Beam Current Measurement**" 25th International Conference on Nuclear Engineering 2017, Shanghai, China.

Hobbies

Playing chess, learning languages, learning musical instruments, and traveling.